

The Internet of Things (IoT) basically involves the collection and management of a large amount of data from a vast network of devices and sensors. The data that has been retrieved is processed and then shared with the other units or devices on the network, so that real-time decisions can be made. An example of such networks is cars monitoring themselves — having the capabilities to ‘know’ what to do in particular instances. Another such example is security alarm systems that are installed in houses. These systems can judge which situations are safe and when the alarm needs to be set off to warn the inmates of the house.

IoT, therefore, needs a program to help it easily connect with different devices, and maintain this connectivity in the ecosystem. The IoT architectural environment is divided into four major sections:

- The sensors that create the data
- The hubs or local gateways that organise it
- The geographical locations
- Centralised servers that collect the data

A basic sensor generally uses C as the programming language, as the latter can work directly with the RAM. For the rest, developers should pick and choose the language that best suits them and the build. As the hubs are like small consoles or smartphones, they will use a standard OS. So communication via a command line will feel the same as what developers have grown accustomed to.

The top five languages for building IoT solutions are:

- Java
- C
- JavaScript
- Python
- PHP

Java

Java is a programming language that offers portability to the network, and developers can learn it easily. These two aspects come together to make Java the perfect program to help devices connect with each other. Almost all devices, starting from PCs to mobile phones, use Java. It is also an integral part of the world of the Internet, which makes it a good choice for IoT. Java offers every device the best level of functionality, giving it a high level of security and a good amount of scalability. Also, the fact that Java has a huge ecosystem makes it even more suitable for IoT. Senior Java J2EE developers are creating innovative applications that will help in achieving the goal of a connected world.

When developers write an embedded application, they need to take lots of factors into consideration. These include the real-time operating system to be used and the various protocols that need to be connected with it. When Java Me is used, it abstracts all these factors. This makes it easier for the written application to run on various devices without any call for change, anywhere.

The benefits of using Java for IoT: Listed below are some of the key advantages of using Java for IoT.

1. It gives an enhanced user experience.
2. It enhances the life cycle of the product.
3. Increases market reach.
4. One can have control over the roadmap and BOM – since different kinds of hardware support Java.
5. The resource availability is high.
6. The support cost is comparatively low.
7. It has been proven to be reliable and secure.

The C language

One of the most important programming languages in the IoT system is C. This is the lowest layer of software, and the one closest to the hardware. C has been the foundation for many other coding languages over the years. Hence, mastering it is a basic necessity for anyone working with IoT projects.

C doesn't require a lot of processing power and is available on almost every advanced embedded system platform. It is procedural rather than object-oriented as it does not have built-in capabilities. This programming language is compiled, making it great for IoT projects.

The syntax is cluttered with punctuation, and there are a million different little mistakes you can make, but the language is still the first choice for many programmers who write for the lowest layer of software. The language hides nothing from you, and that means you can fiddle with every part of the code to squeeze out the best performance from an underpowered device. Every bit can be flipped. Every value on the stack is available. Just don't make a mistake, because there are only a few safety nets.

The benefits of using C for IoT: Here are some of the advantages of using C for IoT.

1. C language is the building block for many other currently known languages.
2. C is a highly portable language.
3. There are only 32 keywords in ANSI C, and its strength lies in its built-in functions.
4. Another important advantage of C is its ability to extend itself.
5. C language is a structured programming language.

Python

Python is mostly used for writing Web applications, but it has gained popularity in the IoT system. It is an interpreted language that offers readability with syntax, but without compromising on the size. This language has a large number of libraries, and can get more done with fewer lines of code. Python's clean syntax is suitable for database arrangement. In case your app needs the data to be arranged in a database format or use tables, Python is the right choice.